Answer Key — Formation of Wind

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

Q1 **(b) high to low pressure** — air flows from high to low pressure.

Q2 **(b) decreases** — high-speed air has reduced pressure.

Q3 **(b) from sea to land** — low pressure forms over the warmer land, so wind blows sea → land.

Q4 **(b) open** — open windows equalise pressure and help keep the roof from blowing off.

Q5 **(a) Both true, R explains A** — the pressure difference lifts a weak roof — exactly as R explains.

Q6 **land** — the land breeze blows land → sea at night.

Q7 **faster** — a bigger pressure difference gives faster wind.

Short answer

Q8 From high pressure to low pressure; the greater the pressure difference, the faster the wind.

Q9 Sea breeze: daytime, low pressure over warm land, wind blows sea → land. Land breeze: night, low pressure over warmer sea, wind blows land → sea.

Q10 Fast wind lowers pressure above the roof; open windows let inside and outside pressure equalise, reducing the lifting force on the roof.

Long / application

Q11 During the day land heats faster than the sea, so air above the land becomes warm, lighter and rises, creating a low-pressure region over land. Cooler, higher-pressure air over the sea flows in to take its place, blowing from sea to land — the sea breeze.

Q12 Blowing creates fast-moving air between the balloons, which is a low-pressure region. The higher surrounding air pressure pushes the balloons inward. Blowing harder makes the air faster, lowering the pressure more, so they approach faster — high-speed air is accompanied by reduced pressure.

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